LOGISTICS AND SUPPORT IN THE DE&S, AND HOW THE FUNCTION IS DELIVERED AND ENABLED BY THE DE&S IN JULY 2012

Introduction

1. Logistics is defined as the science of planning and carrying out the movement and maintenance of forces[1]. Its purpose is to maximise the freedom of action for operational commanders by giving them confidence that the right support will be delivered when needed. In its most comprehensive sense, Logistics comprises those aspects of military operations that deal with:

- Design and development, purchasing, storage, supply, distribution, movement, asset management, tracking, in-service support, maintenance, recovery and disposal of materiel.
- Transport of personnel.
- Purchasing or construction, maintenance, operation and disposition of facilities.
- Purchasing or furnishing of services.
- Medical and Health service support.

2. The interpretation of what is Logistics varies widely in both the military and commercial spheres. The use of the term Logistics can also be synonymous with Support. In the context of MoD Equipment Acquisition, Support is the function of sustaining a capability through-life, such as a ship, an aircraft or tank, through a combination of Supply, Support Engineering, and Support Services[2], which are enabled by Logistics Information.

3. There are clearly similarities with Logistics and Support in the commercial sector, but the military environment does have some unique characteristics. The Armed Forces give the means to use or threaten force as an instrument of National power. To sustain this capability the operational commander must be assured that the equipment will work and that resources will be replenished. To fail in the delivery of Logistics and Support means to risk the lives of UK nationals or other Allied personnel. Whilst every situation may not be as extreme as warfighting, Support to the Armed Forces must be responsive, agile, innovative, dependable and have global reach[3]. Lines of communication are typically worldwide, complex and fragile[4]. Where Business drivers are financial, large inventory stocks are typically considered to be bad practice and the supply chain is optimised to Just-in-Time. However, in the military context where demand is often unpredictable or must satisfy contingency, Support and Supply must meet every eventuality from Just-in-Time to holding stocks for the Just-in-Case. Similarly the equipment range is immense, with technology that is often cutting-edge and used in extreme ways and conditions. Demand might be unforeseen, such as Urgent Operational Requirements, through to equipments that remain in-service for many years if not decades[5].

4. **Aim**. The aim of this paper is to describe the function of Logistics and Support, and how this is delivered and enabled by the DE&S as at July 2012.

5. **Scope**. For the purposes of this paper the following are out of scope: Information Systems & Services (ISS), Naval Bases and support to Other Government Departments.

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[1] JDP 4-00 Logistics for Joint Operations.
[2] In-house services or contracted service provision arrangements e.g. Contractor Logistics Support (CLS).
[3] Operational scenarios are defined by the Defence Military Tasks: they range from Nuclear deterrent, direct intervention through to Civil Contingency e.g. fireman's strike, flood response or Op OLYMPICS.
[4] The closure of the Pakistan/Afghanistan border on the ground Line of Communication (LoC) had an effect upon the supply chain to UK forces on Op HERRICK.
[5] One example is Obsolescence Management. Most MoD helicopters have or are expected to stay in service for more than 40 years; the FV430 series armoured personnel carrier entered service in 1960s; and Type 45 Destroyer class will be in-service for 30+ years.

Logistics Process

6. Logistics in the military context is an End-to-End (E2E) process from Industry through to the frontline in an operational theatre, or to a deployed Force Element such as a Royal Navy ship. It stretches across a network of nodes with multiple processes, through which materiel flows and services are provided, with many interfaces between stakeholders who include the Commands[6], the Permanent Joint Headquarters (PJHQ), MoD Head Office, DE&S and Industry.

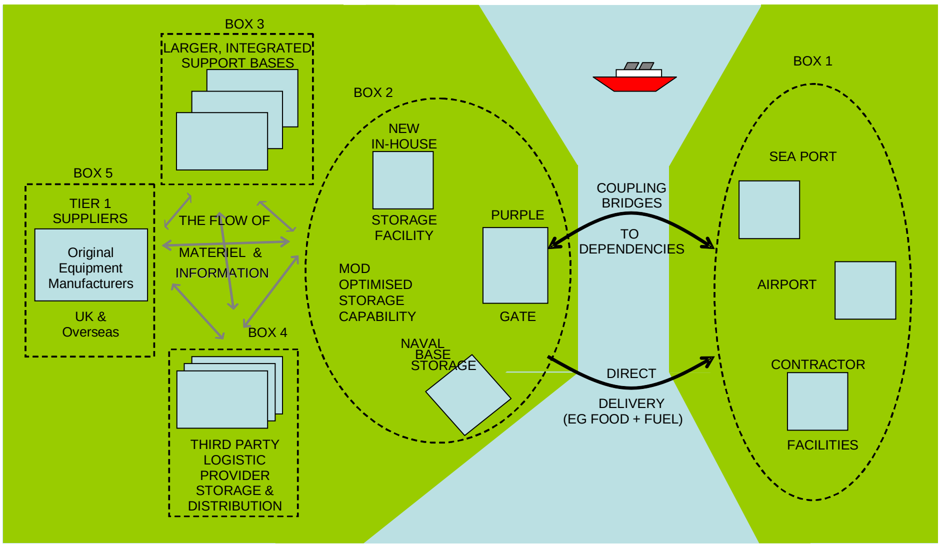

Figure 1 – The Support Chain Landscape Schematic (5 Box Model)

7. By way of example, Figure 1 is a schematic of the typical Support Chain landscape in the Land domain. In terms of Equipment Capability, the MoD and the Commands define the military User Requirement and the DE&S then delivers the specified capabilities. In order to sustain the soldier, sailor or airman, and their respective equipment, the DE&S must provide the necessary resources, consumables and services to maintain the specified levels of operational capability and availability. The 5 boxes represent the generic network through which information must flow specifying demand and performance, materiel must flow to and from the operational theatre[7], and support services must be provided to maintain the Force.

8. Military Capability is achieved by integrating the constituent elements: People, Equipment, Doctrine, Training, Information, Logistics, Organisation and Infrastructure. These elements are called the Defence Lines of Development (DLODs). The DE&S role in both the Equipment and Logistics Lines of Development is key from a Support perspective, as well as an element of ‘Logistics and Support’ to each of the other Lines of Development. Within the Levene Model, the DE&S provides the Acquire or Acquisition function[8]. In the current operating model where the DE&S is an integral part of the MoD, the DE&S conducts Logistics and Support roles as; a Requester, by defining elements of the military User Requirement on behalf of the Commands; as a Deliverer, it translates the User Requirement into optimised solutions; and as an Executer, it physically conducts certain elements of the Logistics and Support function e.g. warehousing and storage.

9. The Chief of Defence Materiel (CDM) as the Logistics Process Owner (LPO) is responsible to the Defence Board for managing the total performance and providing

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[6] The four Commands are the Single Services - Navy, Army and Air, and Joint Forces Command.
[7] The Purple Gate is a node to regulate the flow of materiel into the Joint Supply Chain across the Coupling Bridge, which is any means of transporting the materiel to an operational theatre i.e. by land, sea or air.
[8] Acquisition is how the MoD works with industry to provide the necessary military capability to meet the needs of the Armed Forces; it covers the setting of requirements; the selection, development and manufacture of a solution to meet the requirements; the introduction into service and support of equipment and other elements of capability through life; and its appropriate disposal.

assurance across the E2E Logistics Process. Therefore CDM is both process owner and Head of DE&S, and as such receives two Letters of Delegation from the Permanent Under Secretary (PUS): one as a Top Level Budget (TLB) Holder and one as the Logistics Process Owner. CDM is empowered to set binding policies, standards and rules necessary to deliver the required Logistics Effect in the most cost effective manner. CDM governs the Logistics Process through the Defence Logistics Board (DLB) according to Defence Strategic Direction (DSD), the Defence Plan and the Defence Logistics Sub-Strategy, which includes the Defence Logistics Direction (DLD). The DLD augments the DSD Planning Assumptions by providing greater depth and detail against those planning guidelines relating to logistics support and sustainability. CDM delegates the routine responsibility as process owner to ACDS (Log Ops). It is ACDS (Log Ops) task to bring coherence and consistency to the development of Defence Logistics Policy, to assure the E2E Logistics Process, and to lead on international operational logistics issues including the promotion of Collective Responsibility[9]. ACDS (Log Ops) governs these activities through a combination of the Defence Logistics Steering Group, the subordinate Defence Logistics Working Group, and the International Logistics Planning Committee. There is a continuous need for strategic logistics advice to MoD Head Office, provided by ACDS (Log Ops), who supports DCDS Mil Strat & Ops through the provision of logistics advice to military strategic planning and strategic command of operations.

10. ACDS (Log Ops) sits within the current DE&S organisational chart. However, the actual role is one of Requestor, setting Logistics and Support policy and direction to the DE&S and the Commands from a central coordinating position. In this role ACDS (Log Ops) is referred to as the Joint Logistics User[10] and is responsible for the conceptual development of the Defence Support Network.

11. CDM provides assurance of the E2E Logistics Process through a combination of the Annual Assurance Report that culminates in his submission to the Defence Audit Committee, and through the DLB risk management process. Logistics Readiness risks (Force Elements@Readiness and Force Elements@Sustainability) are reported by the Commands, TLB holders and Trading Funds through their Capability, Operations, Standing Tasks and Recuperation inputs to the Defence Quarterly Performance and Risk Report. ACDS (Log Ops) draws on these inputs to produce a Quarterly Assessment of Sustainability Risk, which provides a coordinated view of sustainability risks and shortfalls against the Force Elements listed as Adaptive, Committed and Responsive forces in the Defence Plan.

Demand

12. Logistics and Support demand is placed upon the DE&S by MoD Head Office, the Single Service Commands, Joint Forces Command and PJHQ. These stakeholders determine the military User Requirement and specify levels of platform and equipment availability to support Force Generation i.e. to maintain Force Elements@Readiness (equipped, trained and ready to deploy), and set the Requirement to sustain these Force Elements for the duration of an operation or campaign i.e. Force Elements@Sustainability. These high level requirements are articulated through the respective Joint Business Agreements (JBAs) between each of the Commands and the DE&S Chiefs of Materiel (COMs), and are the mechanism by which the COMs are held to account for the DE&S Support outputs[11].

13. Almost every Business Unit across the DE&S has a part in satisfying the Logistics demand as part of the Logistics and Support function. Success is achieved through the integration of each functional discipline; Project Management, Financial, Commercial, Engineering, International and Information, to design, develop and sustain effective through-life Logistics and Support solutions. The demand is satisfied by the *Delivery* Operating

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[9] Collective Responsibility (CR) in NATO is the move from a national to a multi-national approach to Operational Logistics.
[10] ACDS (Log Ops) has a delegation from Commander Joint Forces Command to act as the Joint Logistics User.
[11] Delivery against the Weapons stockpile liability is conducted outside of the annual JBA process, however, reporting is aligned with the JBAs for coherence. Formally, reporting against the stockpile liability is conducted quarterly against the Defence Armoury Report.

Centres, segmented by domain and capability, which translate the requirement into detailed Support and Supply solutions and then manage the actual supply of equipment, services and commodities from Industry. Other elements are also essential to design and enable the E2E Logistics Process. These *Enabling* Operating Centres include Joint Support Chain (JSC), Technical, and Logistics Commodities and Services (LCS), which provide the policy, guidance, assurance, Support services and Logistics Information Systems. Oversight of Support and Movement to meet the operational needs in both the UK and overseas is the responsibility of Defence Support Chain Operations and Movements (DSCOM), which provides DE&S with the focus for operational planning, management and control.

14. The JBAs for the Maritime, Land and Air domains are managed by three DE&S Customer Support Teams (CSTs) on behalf of their respective COMs. The CSTs provide intelligent advocacy for the Commands' requirements across each single Service and the Joint domain. As well as providing liaison and coordination between the DE&S Operating Centres, the CSTs provide coherent pan domain DE&S representation and input into the Commands' management and operational command groups. The CSTs are also pivotal in providing the Commands with the assurance that their respective intents are being delivered in a safe, professional and efficient manner. Specifically the CSTs provide:

- Operational Support coordination and conduct assurance activity on behalf of their COM e.g. CST (Fleet) delivers a Command facing 24/7 single point of contact into PJHQ and Navy Command to co-ordinate the complex delivery of DE&S support to the continually deployed Maritime Force Elements.

- Conduct Programme and Portfolio Coordination to bring together aspects of change programmes within their respective domains to ensure strategic coherence.

- Act as their COM's Executive Officer to provide a link between customer and supplier, to monitor and challenge performance in both organisations, and to discharge the COMs' wider corporate responsibilities.

15. The fourth DE&S domain is Joint Enablers (JE), which does not have a specific CST. The COM (Joint Enablers) commits to a JBA between the Operating Centres in the Joint Enablers domain and each of the Commands.

Deliver

16. The responsibility for developing, delivering and optimising project and programme Logistics and Support to meet the Customers' agreed performance, time and cost requirements is delegated to each 2* Director in charge of a *Delivery* Operating Centre. These are as follows:

- COM (Fleet) – Submarines, Ships and Naval Bases.
- COM (Land) – Land, Weapons, [JSC and LCS]*.
- COM (Air) – Combat Air and Air Support.
- COM (JE) – Helicopters, ISTAR and ISS.

* *Enabling* OCs

Each Operating Centre is a grouping of Project Teams, which conduct the detailed management for their assigned platform or capability. The Operating Centres through their Project Teams provide the principal interface with Industry both in the UK and overseas. Project Team Leaders are accountable for ensuring that their equipment is safe, fit for purpose and sustainable.

17. Within and around a *Delivery* Operating Centre, there are a variety of staff inputs that are essential for the development and management of effective Logistics and Support solutions. It is not just the responsibility of specialist Logistics staff, but the combined contribution made through awareness and understanding, oversight, professional Subject

Matter Expertise and practitioner skills and experience. Figure 2 is a simplistic representation of the generic contributions in the management of through-life Logistics and Support.

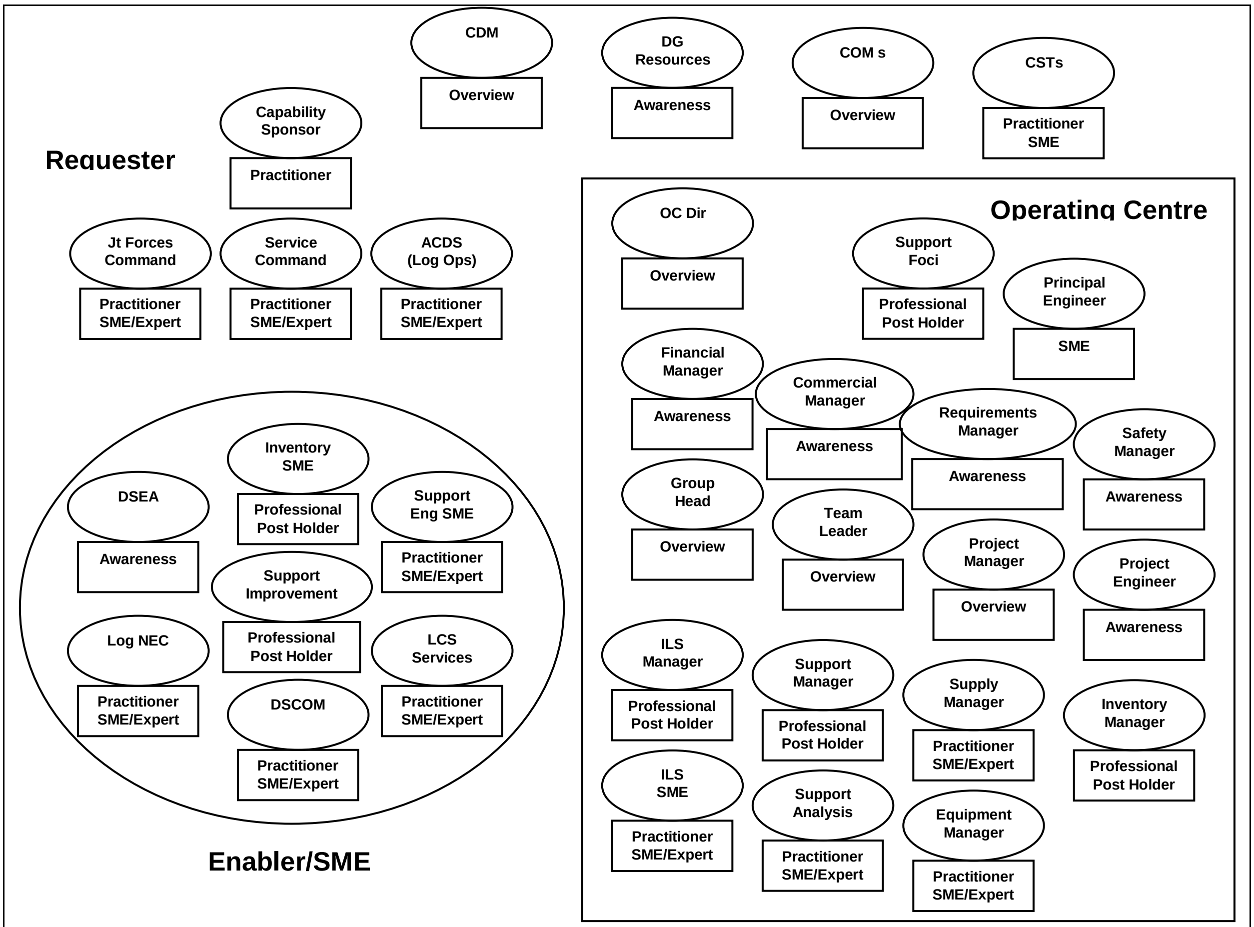

Figure 2 – A generic representation of the contributions towards the management of Logistics and Support in a Delivery Operating Centre

18. Policy and guidance for the development of coherent Logistics and Support solutions stem from the Defence Support Requirements[12], the Acquisition Operating Framework (AOF), JSP 886 – the Defence Logistics Support Chain Manual, and the Support Solutions Envelope (SSE). The Support Solutions Envelope is a DE&S tool that provides a framework of the elements that must be considered to develop and sustain effective Logistics. The Support Solutions Envelope framework is at Annex B. It provides specific guidance to Project Teams; it signposts the array of Defence policies and Technical standards, and is a framework with which to assess the development and effectiveness of a proposed Solution[13].

19. It is mandated that Project Teams will use the Integrated Logistics Support (ILS) methodology[14] to develop Through-Life Support Solutions. ILS is a widely-recognised disciplined approach to managing Whole Life Support Costs that affect a customer and its suppliers. The aim is to influence the evolving System design in order to optimise its supportability. The ILS methodology, which is complementary to the elements in the Support Solutions Envelope, is applied to a project or programme throughout its lifecycle or CADMID phases (Concept, Assessment, Development, Manufacture, In-Service and Disposal)[15]. An indicative example of the ILS activities across the CADMID cycle is shown at Annex C.

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[12] ACDS (Log Ops) mandates compliance with Defence Support Requirements to drive coherence and commonality amongst platform Support Solutions.
[13] Support Solution assessment can be conducted using the SSE Support Solution Development Tool (SSDT), which provides a means of audit through objective evidence.
[14] The methodology is explained in JSP 886 and the Defence Standard 00-600.
[15] For a Service Provision arrangement, it is modified to CADMIT to reflect **M**igration to a service contract and **T**ermination.

20. For each project or capability, the Project Team will determine the most appropriate Support Solution model based upon the User's specified capability and availability requirements. Aligned to this the appropriate Contracting model is determined using the Support Options Matrix[16], which balances opportunity, capability and incentives with the level of cost and performance risk. The DE&S approach is to move away from transactional responsibilities that can be delivered better and more cost effectively by Industry, and concentrate on performance management and decision-making: this theme is known as the shift from Provider to Decider. In terms of Logistics and Support, the trend is towards passing more Support responsibility to a Contractor, moving away from traditional in-house Support arrangements towards Contracting for Availability and Contracting for Capability. By way of example different Support Solution models for the Maritime domain are at Annex D. This is equally applicable in the Air domain where over the past 10-15 years the majority of aircraft Support has migrated towards varying forms of service provision, and the Land domain is now increasingly looking to adopt similar Strategic Support Supplier arrangements.

21. Support strategy planning can start as early as the Concept Phase and matures during the Assessment Phase as ILS and Engineering activities shape the capability design. In the early stages of Acquisition, the Support strategy is based upon assumptions, experience and modelling. In determining specific Support Requirements, the Project Team will be guided by the respective Domain's operating, support and maintenance doctrine, which defines the types and levels of Support and Repair. Design activities inform the inventory requirement and modelling helps determine the inventory range, scale, storage and the optimum disposition across the E2E supply chain. Each Project Team is required to have a comprehensive Inventory Plan.

22. Inventory optimisation and Support improvement are through-life activities. It is a well-established lesson that at least two thirds of a platform's Whole Life Cost is as a result of Support cost during the in-service phase. Post Main Gate approval the key milestone is the Logistics Support Date, which is part of User acceptance where all aspects of the required Support Solution are assessed to be in place. From this point forward, the Project Team manages the equipment in-service.

23. Depending on the Support Solution model, Project Teams will play a greater or lesser part in providing in-service Support. This may range from providing real time support advice to units on deployed operations, controlling the maintenance policy for individual items of inventory, addressing issues that arise from equipment safety, failure or reliability, through to managing and disposing of inventory. The Project Teams manage or deliver all of the services and processes defined in Annex B across the Key Support Areas (KSAs) and Governing Policies (GP).

24. One key initiative that is underway across Defence is to reduce the size of the Defence Inventory and improve Inventory Management practice. The Defence Inventory has grown to such an extent that it is no longer sustainable either physically or financially. Therefore significant effort is being focused towards reducing the value, quantity and volume to meet the future Defence requirement.

Design

25. CDM has tasked COM (Land) with the delivery of an effective and efficient Joint Support Chain, part of which is the design and optimisation that is enabled by the JSC Operating Centre. Director JSC is responsible for the oversight and function of the Joint Support Chain, to optimise E2E support chain policy, process and performance, to coordinate and facilitate strategic movement and operational logistic support arrangements, and implement Logistic IS improvement programmes across Defence.

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[16] As specified in JSP 886 with guidance provided by DTech ESCIT.

26. **Support Chain Management (SCM)**. SCM is a Business Unit within the JSC Operating Centre and provides the strategic planning across the Joint Support Chain, and a management, integration and planning capability to increase efficiency, Value for Money, and the coordination between stakeholders: Operating Centres, Project Teams, ACDS (Log Ops), PJHQ, the Commands and Industry. SCM's role is to maintain and develop the Joint Support Chain by: maintaining and developing Joint Support Chain policy (including Supply Chain, Transport, Movements and Through Life Support policy); managing the overall User requirement for JSC outputs and the requirement setting for the LCS Operating Centre; providing advice and guidance to Project Teams on the development of effective and coherent Support Solutions, compliance with Support Policy and Inventory Optimisation; measuring the performance of the Joint Support Chain; and providing specialised support services including NATO codification[17]. Head SCM supports Director JSC in his dual roles, as CDM's Support Champion for Defence, by maintaining and developing the Support Solutions Envelope, and in his role as Logistic Skills Champion, through the development and provision of effective joint individual training, education and other interventions.

Deploy and Recover

27. Defence Support Chain Operations and Movements (DSCOM) is a Business Unit within the JSC Operating Centre. In its strategic role it provides CDM with an Operations Centre to enable DE&S to support global Defence activity. It coordinates Joint Support Chain activity and movements support to operations, the Defence Exercise Programme, deployed Naval platforms, and the Permanent Joint Operating Bases. DSCOM is the lead for delivering the DE&S operational battle rhythm and provides the critical link between DE&S Project Teams and deployed force elements to ensure the timely and efficient flow of materiel and support, and enables Defence to realise the synergies between its organic and commercial multi-modal transport options.

28. DSCOM has a three pillar structure organised into Operational Plans, Operations Centre and Defence Movements[18]. DSCOM acts as the lead DE&S planning interface with the MoD, PJHQ and the Commands. The primary focus in support of current operations is through PJHQ. Outputs to the Commands are bounded by the JBAs within the funded programme. Specifically; the Operational Plans section coordinates DE&S planning in support of contingent and deliberate operations and major exercises; the Operations section enables the functionality of DSCOM and provides the focus for the coordination of effective DE&S support to operations, training and crisis management worldwide, reacting to short notice change in the physical delivery and tracking of materiel; and the Defence Movements section conducts the acquisition and planning of strategic lift in support of the authorised Statements of Movement Requirement and contributes to the production of Load Allocation Tables for surface and air movements.

29. Strategic lift is provided by a mix of military transport and where shortfalls emerge, contracted civilian assets. These include Air and Sea Charter and a Private Finance Initiative for RORO[19] sealift operations worldwide. Routine sustainment freight is moved under the Global Freight Transport Services Contract. DSCOM also sets contracts for bespoke Lines of Communication (LoC) such as the Southern LoC to Afghanistan, which combines a weekly commercial liner service with contracted road transport, and the Northern LoC to Afghanistan that involves sealift, rail and road contracts to move materiel into theatre. All contracts are set, operated and monitored by DSCOM.

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[17] Codification means the application of unique identification and classification of Items of Supply, using a common supply language. This information is recorded to distinguish unique Items of Supply using NATO Stock Numbers (NSN), to enable national and international logistic support, data management, supply and inventory management, throughout the life of an equipment, by the UK MOD and its NATO partners.

[18] DSCOM has 135 staff: 2/3 military drawn from all three Services and 1/3 Civil Servant. A large proportion of the military manpower are Movement Specialists. Commercial and financial staff support DSCOM's outputs and a small number of commercial embeds provide the global freight business with specialist support.

[19] RORO are Roll-On Roll-Off ships.

Logistics Information

30. Logistics information is a critical enabler that underpins the management of Logistics and Support at all levels, and the optimisation and performance of the E2E Logistics Process. Log NEC is a Business Unit within the JSC Operating Centre. The aim of the Log NEC programme is to deliver step change improvements in the MoD's Logistics Networked Enabled Capability through the implementation of new projects[20], enabling the rationalisation of current logistics processes and information systems. It is an IS-enabled business change programme.

31. The Log NEC organisation is configured to address three core activities:

- **Sustain** existing logistics information services that underpin operational capability. This includes logistics processes, information and integration services, applications and infrastructure for Supply and Through Life Engineering Support. The scope covers Support to the whole range of equipments e.g. surface ships, submarines, fast jets, fixed wing aircraft, rotary wing aircraft and approximately 60,000 different equipments in the Land environment ranging from Apache helicopters to vehicles.
- **Improve** service availability and business continuity, support network performance, equipment support through better diagnostic and prognostic capabilities, decision support, and data and integration services to improve data quality, agility and more responsive integration of new capabilities into the Defence Support Network to meet Industry demand for future contracting models.
- **Transform** current capabilities to deliver IS-enabled business and operational processes in asset management and inventory management, in order to provide competitive advantage for operational commanders and to optimise the E2E Logistics Process.

32. The Future Logistic Information Service (FLIS) is central to the Log NEC programme. The project has established Boeing Defence UK (BDUK) as the strategic delivery partner, who took over responsibility for the delivery of logistics information services, driving out cost in the programme by acting as a single commercial interface with over 50 suppliers, and managing the delivery of over 200 logistics applications. Through its incentivised contract BDUK is delivering sustained and optimised information services and capabilities, and through innovation and transformation it will radically improve the current IS architectural paradigm and quality of Logistics data that has been notoriously poor in the past.

33. Three major change programmes already underway are:

- **JAMES**. The Joint Asset Management and Engineering System is a Management Information System that provides global and near real-time visibility of asset and engineering data for Land-managed equipments (rollout scheduled to complete in 2014).

- **MJDI**. The Management of Joint Deployed Inventory (MJDI) is a Management Information System that gives the User a complete view of the entire Defence Inventory, wherever it is held and across all three Services. All Users will follow the same ways of working, using the same application on a common IT system, which will enable demanding, receiving, supplying and maintaining details of items ranging from clothing and rations, armoured vehicles, aircraft and ship spares to helicopter parts, munitions and fuel (rollout scheduled to complete in 2014).

- **BIWMS**. The Base Inventory & Warehouse Management Services (BIWMS) will provide a common Management Information System to transform the existing single

[20] These are business-led projects aimed at improving the processes and information systems underpinning asset management and asset tracking.

Service logistics processes into common Joint processes for Base inventory management, Warehouse management, and provisioning and procurement improvement (rollout scheduled to complete in 2015). The system should also replace current bespoke specialist applications that manage many of the legislative requirements for munitions storage, transport and through-life monitoring (including at sub-component level).

34. **Logistics Information Strategy**. Log NEC direction is provided by the Capability Sponsor, Cap C4ISR and ACDS (Log Ops), and further strategic direction is included within the ACDS (Log Ops) Defence Logistics Information Strategy and the MoD's Chief Information Officer's Information Strategy.

35. **Industry Engagement**. The Log NEC capabilities must reach back into the UK industrial base. As a result, these IS-enabled business change programmes require process change that may impact on Industry partners. This is particularly the case where extant service provision arrangements specify information exchange requirements shaped around the legacy processes. Log NEC is working with Industry through the 'Green Box'[21], a jointly funded project that analyses the effect of MOD process change on Industry, and works with Project Teams and their Industry partners to bring about timely contract change. Further Logistic information exchange requirements are addressed through the Joint Information Group working with the UK Council for Electronic Business (UKCeB), which represents many small and medium size Enterprises.

Logistics Commodities and Services (LCS)

36. The functions in the LCS Operating Centre can be segmented into three main types of activity: Logistics Services, Logistics Commodities, and Development and Change. The Operating Centre was specifically formed from elements of the JSC to pursue outsourcing of its activity. Its primary purpose is to ensure the continuity of military supply, which is essentially an Execute function. Whilst specific demands are placed upon LCS by the Commands, Operating Centres and Project Teams, the central Customer role is managed through the SCM Business Unit. Once outsourcing is complete, the DE&S Main Board intent is for the residual LCS elements and JSC to be combined into a single Operating Centre.

37. Firstly, Logistics Services handles the physical activity of receipt, storage, distribution, disposals and related activities for all products within the home and non-deployed base. The main output is the storage of non-explosive (i.e. not munitions) products and movement of all products (including munitions) within the home base (UK & Germany). The products include not just those controlled by Logistics Commodities but also the inventory controlled by all other Project Teams within DE&S. Logistics Services operates specialised storage for small arms weapons, classified materiel and hazardous stores including fuels and oils, as well as more general warehousing for the MOD's general stocks, ranging from uniforms to spare parts for ships and aircraft. In all, it employs nearly 2000 staff across more than 20 sites in UK and Germany, in more than 60 warehouses. Logistics Services operate the Purple Gate, which is a 24/7 process to ensure the regulation of materiel flow into the Joint Supply Chain across the Coupling Bridge to the worldwide operational theatres. It provides control, coordination, consolidation and tracking of consignments to the deployed forces. Other services also include: delivery of a postal and classified courier service; and the Disposal Services Authority (DSA), which manages the disposal (by sale ideally) of surplus MOD assets and generates significant revenue for the Department.

38. Secondly, Logistics Commodities handles the procurement, inventory management and supplier management for Clothing (operational and working dress, personal protective clothing, parade and ceremonial uniforms), Fuel (bulk fuels, oils, lubricants and gases), Food

[21] Green Box is a grouping of Log NEC and Industry representatives, which identify potential issues, use architectural methods to define the requirement and activities that are required for resolution, identify the organisational stakeholders, and facilitate the resultant action teams.

(fresh, ambient, cold, in-flight and operational ration packs), Medical Supplies (equipment, consumables, pharmaceuticals, blood products and Medical Countermeasures), General Supplies (non-platform specific goods) and Movement Support Services (freight forwarding, business travel, global removals and family relocation service). Responding to requirements set in JBAs with the Commands and PJHQ, Logistic Commodities manages over 400 contracts for goods and services with around 300 different suppliers and c50,000 live NATO Stock Numbers to ensure continuity of supply to customers across Defence in the UK, in overseas bases and in operational and training theatres around the world. Except for the Food Vote, which is currently managed by Logistic Commodities, funding for the commodities and services is held by the User TLB Holders with payment on consumption or claimed against the Net Additional Cost on Military Operations (NACMO). All contracts comply fully with EU and national legislation to ensure minimal risk of litigation and to uphold the MOD reputation with Industry.

39. Thirdly, Development and Change, which handles all the major changes within the Operating Centre, but ostensibly at this time it is the transformational project seeking to outsource the majority of LCS activity to the commercial sector. The LCS Transformation Project is a Category A project that has received Initial Gate approval to enter into the Assessment Phase. Potentially 85% of the existing LCS staff could be outsourced to Industry in 2015 with a projected saving of £1Bn over the next 20 years. The project sits within a programme of transformation including fuel procurement and distribution, the World Wide Food Services project, and new options for postal and courier services and disposal activity.

Weapons Sector

40. Defence Munitions (DM) is mandated to receive, store, maintain and issue the range of General Munitions and Complex Weapons that are directed by the MoD and procured through the Project Teams to equip the Armed Forces. In this role DM is the custodian and provider of all in-service support to the UK's stockpile of munitions and related non-explosive items. It is the provider of niche engineering services, including but not limited to, Naval Gun Work, Exercise Mines, Handling Gear and Towed Array. In all, DM employs nearly 1200 staff across 9 sites in UK and Germany[22].

41. Processing is a complex and expert task that involves the following:

a. General Munitions - the examination, repair, pre-issue inspection, safety inspection and preparation of stock for disposal.

b. Complex Weapons - the assembly, integration, test (often under contract), examination, repair, maintenance, modification, configuration management and preparation of stock for disposal.

This is all conducted in specialised and bespoke facilities across the DM sites. The wrap around to this is a Permissions and Restrictions team in Head Office, which provide written approval for all explosive and non-explosive activity, and provide safety assurance to the relevant Inspector of Explosives.

42. The disposal of munitions is managed by the relevant Project Team, which delivers a solution usually by contract, for disposal either through the Long Term Partnering Agreement with QinetiQ (e.g. small arms ammunition) or by individual contracts on a case-by-case basis. This would usually involve DM preparing stock for disposal e.g. to defuze Barmine before transportation to a disposal site.

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[22] The activity equates to nearly 100K Units of Space (each UOS is 1.93m^3) and £5.5 Bn of stock with nearly 50K issues and 30K receipt transactions per year.

43. The Weapons sector is undergoing a transformation programme to improve business processes to meet future demand with stockpile optimisation, better planning, late configuration of missiles, and where possible to provide direct delivery to the Commands.

Logistics People

44. Director JSC is the Logistics Skills Champion for MoD and Logistics Skills Director for the DE&S. The role provides a functional, strategic overview of business trends and critical issues across the Logistics job family. It includes the responsibility to enhance professionalism, provide early notification of key corporate workforce or skills needs, and to shape appropriate Corporate HR interventions to meet these functional needs. Director JSC discharges these responsibilities primarily through the Job Code Sponsors for the following 11 job codes, which make up the Logistics job family:

- Afloat Support Group
- Configuration Management
- Driving
- Through Life Support & Engineering
- Supply Chain Management
- Materiel Accounting
- Port Operations
- Postal & Courier
- Salvage & Marine Operations
- Storekeeping & Warehousing Ops
- Transport & Movement

45. The Logistics job family includes both military (20%) and civilian (80%) personnel comprising in excess of 10,400 staff across all TLBs, of which 50% are within DE&S. Within the DE&S, the function of Logistics and Support is conducted by military, Civil Service and civilian personnel. Whilst many of the appointments are interchangeable between military or civilian staff, the military component brings practical experience of the operational context, whilst the civilian staff tend to bring greater continuity and understanding of the Industrial and commercial environment, and specialist subject matter knowledge. The Through Life Support & Engineering (TLS&E - previously ILS and ETS) and the Supply Chain Management (including Inventory Management) jobs are identified as key functions for upskilling in DE&S, as approximately 97% of the personnel in these job codes are within the DE&S and the functions are considered to be core business. It should be noted that, unlike the Commercial and Finance functions, the Logistics Skills role does not currently have any grade/post management authority for any of the job codes.

46. Operating Centre personnel in other job families/codes e.g. Engineering, Programme and Project Management, and Business Support are also often actively involved in and influential to Logistics and Support function, particularly for TLS&E and Supply Chain Management.

47. **Training and Education (T&E)**. As Logistics Skills Champion, Director JSC is the Training Requirements Authority for Joint Individual Logistics T&E in accordance with the policy described in JSP 896 – The Defence Logistics Training & Education Handbook. A key role of the Skills Champion is to identify, promote and develop suitable functional T&E for those within the job family and also for those in other functions who need Logistics knowledge and skills. A formal T&E strategy for Logistics was published in 2010. Internal licences are mandated for those in the TLS&E and Supply Chain Management job codes. Those who wish to develop a career in Logistics are encouraged to consider studying for a relevant professional qualification. Annex E contains a list of the Logistics T&E that is endorsed and/or sponsored by the Logistics Skills Champion.

48. **Upskilling Budget**. In addition to the funds allocated by Business Units and Project Teams for staff specific training, a DE&S Upskilling Programme fund is allocated to Skills Directors to fund specific functional training, development interventions and initiatives. For FY 12/13 the Logistics Skills allocation is £500K.

Logistics Infrastructure

49. Infrastructure is a key enabling resource for the effective delivery of Logistics and Support. The DE&S has a presence at over 120 locations in the UK and overseas, which includes MOD sites, Main Operating Bases and collocations with Industry partners. Not all of this DE&S estate is related to Logistics and Support.

50. The major elements include the DE&S-owned Naval Bases and outstations including the Sea Mounting Centre at Marchwood, which is the UK's Sea Port of Embarkation; the UK maritime Oil Fuel Depots, the Government [Fuel] Pipeline and Storage System and the West Moors Petroleum Deport, all of which are operated through the Oil Pipeline Agency (OPA); the Defence Munitions sites including maritime ammunition berths; and the LCS non-explosive storage sites in UK and Germany.

Annexes:

A. Common Policies and Guidance for Logistics and Support.
B. Support Solutions Envelope (SSE).
C. Indicative View of Integrated Logistics Support activities across the CADMID Cycle.
D. Examples of Service Provision Support Arrangements.
E. Logistics Training and Education.

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COMMON POLICIES AND GUIDANCE FOR LOGISTICS AND SUPPORT

By way of example, the following is a non-exhaustive list of publications and policies that act as direction and guidance for Logistics and Support staff:

DEFENCE POLICY

Defence Strategic Direction (DSD) 2011
Defence Plan 2011
Defence Logistics Sub-Strategy 2011, Parts 1 – 4
Defence Logistics Information Strategy 2011, Parts 1 - 2

POLICY, GUIDANCE AND LEGISLATION

DSCOM

JDP 4-00 – Logistics for Joint Operations
JSP 327 – Manual of Movements
JSP 751 – Joint Casualty and Compassionate Policy and Procedures
JSP 800 – Vol 4a: Dangerous Goods by Air
JSP 800 – Vol 4b: Dangerous Goods by Road, Rail and Sea
JSP 800 – Vol 6: Policy for the Management and Use of ISO Containers within the MoD
JSP 800 – Vol 2 Edition 3 (Passenger Travel Instructions)
JSP 886 – The Defence Logistics Support Chain Manual (Inc JSP 336 and JSP 586)
DIN 2011DIN04-046 Provision of a Road Freight Service within UK and Continental Europe
DIN 2010DIN04-134 Provision of a Rail Freight Distribution Service within the UK and between the UK and Continental Europe
DIN 2010DIN04-180 Provision of a Surface Global Freight Transportation Service between the UK and destinations Worldwide
Germany Support Command Standing Orders
Op-Ex Mounting Instruction

AIR LOGISTIC SUPPORT

JSP 317 – Joint Service Safety Regulations for the Storage and Handling of Fuels and Lubricants
JSP 319 – Joint Service Safety Regulations for the Storage, Handling and Use of Gases
JSP 367 – Defence Postal and Courier Services
JSP 454 – Procedures for Land Systems Equipment Safety Assurance
JSP 456 – Defence Catering Manual
JSP 483 – Defence Nuclear Material Logistic Movement and Associated Nuclear Accident Response
JSP 520 – Ordnance, Munitions and Explosive Safety
JSP 543 – Defence Technical Documentation Policy
JSP 567 – Contractor Support to Operations
JSP 568 – Defence Logistics Operational Assurance Handbook (Vol 3 only)
JSP 762 – Weapons and Munitions Through-Life Capability
JSP 790 – Defence Rail Safety Management
JSP 800 – Defence Movements and Transport Regulations
JSP 886 – Defence Logistics Support Chain Manual
JSP 896 – Defence Logistics Training and Education Handbook
JSP 899 – Logistics Process Roles and Responsibilities
JSP 902 – Defence Logistics Sustainability

WEAPONS OPERATING CENTRE

JSP 309 - Fuels and Gases Safety Assurance
JSP 340 - Medical Materiel
JSP 375 - Health and Safety Handbook
JSP 384 - Management of DAS
JSP 392 - Radiation Safety Handbook
JSP 400 - Disclosure of Information
JSP 411 - CBRN Guide for Protected Buildings
JSP 418 - Corporate Environmental Protection Manual
JSP 422 - Tri-service Ammunition Packaging, configuration and Statistic Data
JSP 426 - MOD Fire Safety Policy
JSP 435 - Defence Estate Management
JSP 440 - Defence Manual of Security
JSP 462 - Financial Management Policy
JSP 472 - Financial Accounting and Reporting
JSP 482 - MOD Explosive Regulations
JSP 498 - MOD Major Accident Control Regulations
JSP 503 - Business Continuity
JSP 515 - Hazardous Stores Information System
JSP 520 - Ordnance, Munitions and Explosive Safety
JSP 553 - Military Air Worthiness
JSP 567 - Contractor Support to Operations
JSP 762 - Weapons and Munitions Through-Life Capability
JSP 800 Vol 4a - Defence Movements & Transport Regulations: Dangerous Air Cargo Regulations
JSP 800 Vol 4b - Defence Movements & Transport Regulations: Transport of Dangerous Goods by Road, Rail and Sea
JSP 815 - Defence Environment & Safety Management
JSP 862 Vol 1&2 - MOD Maritime Explosives Regulations: Surface Ships & Submarines
Statutory Instrument 1991/1531 - The Control of Explosives Regulations 1991

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Support Solutions Envelope

KSA1 – Log Readiness & Sustainability

Owner: ACDS (Log Ops)

GP 1.1 Logistics Readiness & Sustainability Requirement

GP 1.2 Contractors on Deployed Operations

GP 1.3 Support Continuity

KSA2 – Supportability Engineering

Owner: Hd SCM

GP 2.1 ILS Process & Support Sol'n Development

GP 2.2 Reliability

GP 2.3 Obsolescence Mgt

GP 2.4 ATS & TME

GP 2.5 Configuration Mgt

GP 2.6 Technical Docs

GP2.7 Disposal

GP2.8 Training & Training Eqpt

GP 2.9 Manpower & Human Factors Integration (HFI)

KSA3 – Supply Chain Management

Owner: Hd SCM

GP 3.1 Supply Chain Mgt

GP 3.2 Codification & Item Ownership

GP 3.3 Inventory Planning

GP 3.4 Packaging & Labelling

GP 3.5 Materiel & Financial Accounting

GP 3.6 Fuels, Lubricants & Industrial Gases

GP 3.7 Eqpt Transportability

KSA4 – Logistic Information

Owner: Hd Log NEC

GP 4.1 Logs Information Requirements Planning

GP 4.2 Information Systems & Services Requirements Planning

GP4.3 Effective Log IS Solutions

Note.

KSA: Key Support Area
GP: Governing Policy

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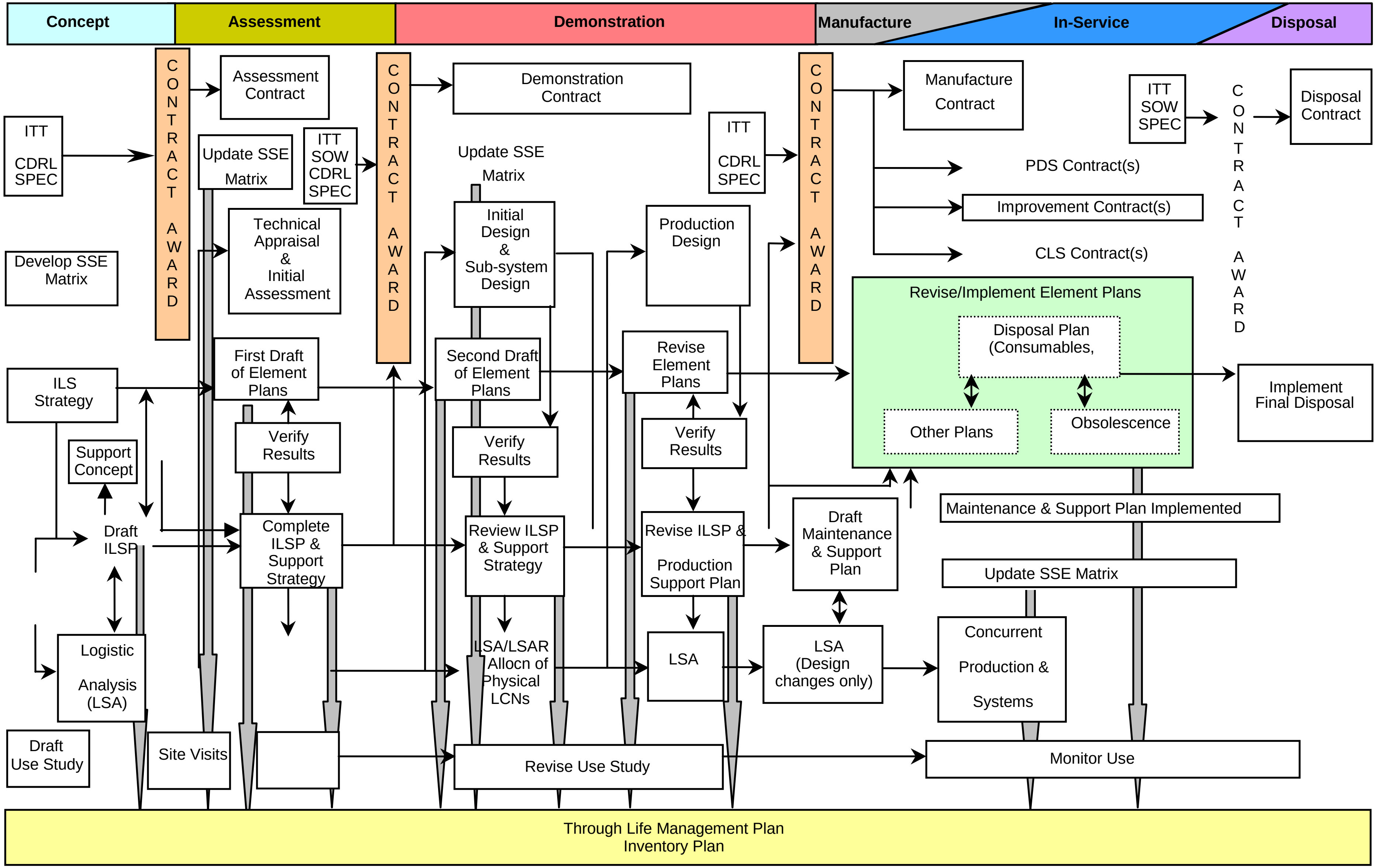
INDICATIVE VIEW OF INTEGRATED LOGISTICS SUPPORT ACTIVITIES



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EXAMPLES OF SERVICE PROVISION SUPPORT ARRANGEMENTS

Ships

- **Maritime Support Delivery Framework (MSDF)**. The MSDF agreements will commercially align Maritime Initiatives under the Maritime Change Programme, namely Surface Ship Support (SSS) and Submarine Enterprise Performance Programme (SEPP), and provide the contractual replacement for the Warship Support Modernisation Initiative (WSMI) Partnering Agreements when they expire in March 2013. There will be two MSDF agreements, one with BAES Surface Ships and one with Babcock Marine. Each Framework Agreement will be underpinned by a number of Annexes, delivering the specific requirements of SSS, SEPP and Naval Bases. MSDF is going forward for Main Gate Approval in October 2012.

- **Maritime Equipment Transformation (MET)**. MET sought to deliver all relevant outputs at their agreed levels whilst reducing administration costs, over three years, consistent with expectations of 14-20% savings from the Administrative Cost Regime (ACR), by realising opportunities to improve Cluster effectiveness and efficiency. The rationale behind the change was that equipment which could be linked by a common procurement strategy (due to the maturity and lower complexity of the equipment) in the near term (1-3 years) would be brought together in MET.

- **Surface Ship Support Alliance (SSSA)**. SSSA is the change programme which aims to transform the way surface ship support is managed in order to deliver the required availability and efficiency improvements whilst sustaining the enterprise through an alliance with industry. See COM below.

- **Class Output Management (COM) for T45, T42, T23 & T22 (QEC likely to align too)**. Under the migration to SSSA management, engineering support is being delegated to COMs. The COM team is an Industry led group that is responsible to the Strategic Class Authority (SCA) for the delivery of the required outputs and for enabling the required savings and benefits. The role of each COM team will be to manage the delivery of support activity on a day-to-day basis and ensure all individual components of support (in both Fleet Time and Upkeep) are properly integrated. The COM Team is the single point of contact between the ship, the SCA and wider Alliance Organisation on day-to-day support issues.

- **Commercial/Leased Ships**. Support will be via a number of varying commercial support/availability contracts/arrangements.

Submarines

- **Submarine Enterprise Performance Programme (SEPP) In-service Support (ISS)**. The MOD in conjunction with its 3 long-term, enduring Tier 1 partners is engaged in a joint change programme in order to sustain the UK's submarine enterprise:
    - Driving sustainability in.
    - Driving performance up.
    - Driving cost down. 3 x Tier 1 suppliers (BAE, Rolls Royce & Babcock).

SEPP comprises 2 principal and balancing components, the collaborative delivery of SUCCESSOR (within the Design and Manufacture phases of the CADMID cycle) and the collaborative delivery of in-service Support that is Flotilla Output Management (FOM).

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LOGISTICS TRAINING AND EDUCATION

Level	Intervention	Provider
Graduate/Post Graduate	Defence Logistics Staff Course - leading to a PG Dip, and on completion of a dissertation, an MSc in Logistics Management	DCLPA in association with University of Lincoln
	PG Dip / PG Cert and MSc for Supportability Engineering	Delivered jointly by Logistics Support Analysts and University of Portsmouth
	Professional education routes via Chartered Institute of Purchasing and Supply (CIPS), Chartered Institute of Logistics and Transport (CILT), and Supply Chain Management (SCM) NVQs, are recommended for staff working in Logistics functions and funded individually subject to a business case	
Internal Functional Licences	Various. Supply Chain Management Development Route map	Defence Academy. Accredited by CILT
	Various. Through Life Support & Engineering Development Route map	Defence Academy
Other*	Defence Strategic Support Management (DSSM)	Defence Academy
	Defence Logistics Management Course	DCLPA
	Supply Chain Management Course	Delivered jointly by Defence Academy and Cranfield University
	Logistic Support Management Course	Delivered jointly by Defence Academy and Cranfield University
	BAEs Supply Chain courses (various)	BAES
	Secondment opportunities for OF5 or above	Arranged through Log Skills
	Logistics mentoring scheme	Arranged through Log Skills

* NB. This table does not include the numerous functional and specialist training courses, which are required for staff within the diverse job codes but which are not directly sponsored by JSC Log Skills.